

Data Modeling: CSCI E-106

Harvard Extension School – Spring 2026

Syllabus

Course Information

Teaching Staff:

Instructor: Dr. Hakan Gogtas

Teaching Assistants:

- Rafael Gomez-Tagle
- Andrea Hatch

Contact Information

Teaching Staff	Contact Information
Hakan Gogtas	hakangogtas@yahoo.com
Rafael GOMEZ-TAGLE	rafael.gomeztagle@gmail.com
Andrea Hatch	andrehatch10@gmail.com

Course Information, cont'd

- Class Dates: Thursdays, January 29-May 16, 5:10pm-7:10pm
 - There will be a 5 min break at 6:05 pm EST
- TA Sessions: Tuesdays at 5:10 pm to 7:10 pm EST
- Office hours are with appointments

- Textbook References:
 - [Optional] Applied Linear Statistical Models, 5th Edition, by Kutner, Nachtsheim, Neter & Li
 - [Optional] Mastering Scientific Computing with R, by Paul Gerrard, Radia M. Johnson
 - [Optional] Linear Models with R, Julian J. Faraway
 - [Optional] Extending the Linear Model with R, Julian J. Faraway

Course Information, cont'd

Date	Class #	Tentative Topic	#	Homework Assigned Date	Homework Due Date
29-Jan	Class 1	Introduction and Simple Linear Regression Models, Part 1	1	30-Jan	6-Feb
5-Feb	Class 2	Simple Linear Regression Models, Part 2	2	6-Feb	13-Feb
12-Feb	Class 3	Simple Linear Regression Models, Part 3	3	13-Feb	20-Feb
19-Feb	Class 4	Multivariate Linear Regression Models		20-Feb	27-Feb
26-Feb	Class 5	Remedial Measures	4	27-Feb	6-Mar
5-Mar	Class 6	Model Validation and Selection	5	6-Mar	13-Mar
12-Mar	Class 7	Polynomial Regressions and Midterm Exam	6	13-Mar	27-Mar
19-Mar	No CLASS - Spring Break				
26-Mar	Class 8	Model Diagnostic Logistic Regression Models Part 1	7	27-Mar	3-Apr
2-Apr	Class 9 Online Midterm Exam will be from Friday, April 3th at 12 pm EST through Wednesday April 8th at 12:00 pm EST.	Logistic Regression Models Part 2 and Poisson Regression Models and Midterm Exam The midterm exam will be available from Friday, April 3th at 12 pm EST through Wednesday April 8th at 12:00 pm EST. You must submit your midterm by Wednesday April 8th at 12:00 pm EST. Once you start the exam, you have to complete the exam in 3 hours or by Wednesday April 8th at 12:00 pm EST.			
9-Apr	Class 10	Regression and Classification Tree Models, Part 1	8	3-Apr	10-Apr
16-Apr	Class 11	Regression and Classification Tree Models, Part 2	9	10-Apr	17-Apr
23-Apr	Class 12	Regression and Classification Tree Models, Part 3	10	17-Apr	24-Apr
30-Apr	Class 13	Machine Learning - Neurol Networks	11	24-Apr	1-May
7-May	Class 14	Autocorrelation in Time Series	12	1-May	8-May
14-May	No Class The Final Exam will be available from May 14th through May 15th. Special Projects are due on May 8th.	The final exam will be available from Thursday, May 14th at 12 pm EST through Friday May 15th at 12:00 pm EST. You must submit your midterm by Friday May 15th at 12:00 pm EST. Once you start the exam, you have to complete the exam in 3 hours or by Friday May 15th at 12:00 pm EST.			

Course Information, TA LAB Session Calendar

Date	Start Time	TA
2/3/2026	5:10 PM EST	Andrea
2/10/2026	5:10 PM EST	Andrea
2/17/2026	5:10 PM EST	Andrea
2/24/2026	5:10 PM EST	Rafael
3/3/2026	5:10 PM EST	Andrea
3/10/2026	5:10 PM EST	Andrea
3/17/2026	No lab	spring break
3/24/2026	5:10 PM EST	Andrea
3/31/2026	5:10 PM EST	Andrea
4/7/2026	5:10 PM EST	Rafael
4/14/2026	5:10 PM EST	Rafael
4/21/2026	5:10 PM EST	Rafael
4/28/2026	5:10 PM EST	Rafael
5/5/2026	5:10 PM EST	Rafael - Prep for Final Exam
5/12/2026	5:10 PM EST	Rafael & Andrea

Course Information, cont'd

Grading:

Assignments	30%
Midterm Exam	30%
Final Exam	30%
Group Assignment	10%

- The two worst homework scores will be dropped
- Homework will be due on Fridays. No Late Homework! No Exceptions!
- Details for the Group Assignment will be shared after the midterm.
- Time management is an important part of your academic success in the Extension school

Computing :

- We will use Posit R Cloud for the class and you will receive a link to register the Posit R for free by the end of this week.
- Create a free account at <http://www.rstudio.cloud> or download the desktop version of rstudio.
- Required to generate PDF reports using the knitr R package for homework, midterm and final exam. (we will go over this in the TA section).

Course Information, cont'd

- Exams will be open books and notes. Further instructions will be provided one week before the exam.
- As always, you are required to follow the Harvard Honor Code.
- Exams will be performed online through Proctorio
 - Proctorio is required for the exam.
 - **Your pc/laptop must have a webcam and microphone (internal or external)!**
 - **You need to have a good internet connection!**
- Midterm Exam Date: **April 3rd** to **April 8th**
- The midterm exam will be available from Friday, April 3rd at 12 pm EST through Wednesday April 8th at 12:00 pm EST. You must submit your midterm by Wednesday April 8th at 12:00 pm EST. Once you start the exam, you must complete the exam in 3 hours or by Wednesday April 8th at 12:00 pm EST.
- Final Exam Date: **May 14th** to **May 15th**
- The final exam will be available from Thursday, May 14th at 12 pm EST through Friday May 15th at 12:00 pm EST. You must submit your midterm by Friday May 15th at 12:00 pm EST. Once you start the exam, you must complete the exam in 3 hours or by Friday May 15th at 12:00 pm EST.
- Special Projects are due on May 8th.

Course Information, cont'd

Suggestions:

- Watch the recording for all classes and TA sections if you can't attend
- TA sessions will be mainly focused on doing exercises in R
- Email your questions if you can't attend the class, we will go over your questions and you can see your questions answered on the recordings
- Time management is important!